Homework assignment for 02YMECH
Chapter: Kinematics

Solutions

1. **Uniform circular motion.**

   A particle moves counter-clockwise on a circle of radius $R = 5$ m with speed $v = 10$ m/s. The center of the circle is at $(x_c, y_c) = (3, 4)$ m. At $t = 0$, the particle is at $(8, 4)$, i.e. 5 units to the right of the center, on the positive x-side.

   (a) *Position vector as a function of time.*

   Angular speed:

   $$\omega = \frac{v}{R} = \frac{10}{5} = 2 \text{ rad/s}.$$

   Because the motion is counter-clockwise and starts at angle 0 relative to the $+x$ direction from the center, we can parametrize:

   $$x(t) = x_c + R\cos(\omega t) = 3 + 5\cos(2t),$$
   $$y(t) = y_c + R\sin(\omega t) = 4 + 5\sin(2t).$$

   Therefore, the position vector is

   $$\mathbf{r}(t) = \big(3 + 5\cos(2t)\big)\,\hat{\imath} + \big(4 + 5\sin(2t)\big)\,\hat{\jmath}.$$

   At $t = 0$,

   $$\mathbf{r}(0) = (3 + 5,\ 4 + 0) = (8, 4),$$

   which matches the given initial condition.

   (b) *Velocity, acceleration at $t = \pi/2$, and perpendicularity.*

   Velocity is time derivative:

   $$\begin{aligned}\mathbf{v}(t) &= \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left[(3 + 5\cos(2t))\,\hat{\imath} + (4 + 5\sin(2t))\,\hat{\jmath}\right] \\ &= (-10\sin(2t))\,\hat{\imath} + (10\cos(2t))\,\hat{\jmath}.\end{aligned}$$

   Its magnitude:

   $$\|\mathbf{v}(t)\| = \sqrt{(-10\sin(2t))^2 + (10\cos(2t))^2} = \sqrt{100(\sin^2(2t) + \cos^2(2t))} = 10 \text{ m/s},$$

   as expected.

   Acceleration is the derivative of velocity:

   $$\begin{aligned}\mathbf{a}(t) &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left[(-10\sin(2t))\,\hat{\imath} + (10\cos(2t))\,\hat{\jmath}\right] \\ &= (-20\cos(2t))\,\hat{\imath} + (-20\sin(2t))\,\hat{\jmath}.\end{aligned}$$

Evaluate at $t = \pi/2$:

$$2t = 2 \cdot \frac{\pi}{2} = \pi, \quad \sin(\pi) = 0, \quad \cos(\pi) = -1.$$

Then

$$\mathbf{v}\left(\frac{\pi}{2}\right) = (-10\sin\pi)\,\hat{\imath} + (10\cos\pi)\,\hat{\jmath} = (0)\,\hat{\imath} + (10\cdot(-1))\,\hat{\jmath} = -10\,\hat{\jmath}\ \text{m/s},$$

$$\mathbf{a}\left(\frac{\pi}{2}\right) = (-20\cos\pi)\,\hat{\imath} + (-20\sin\pi)\,\hat{\jmath} = (-20\cdot(-1))\,\hat{\imath} + (-20\cdot 0)\,\hat{\jmath} = 20\,\hat{\imath}\ \text{m/s}^2.$$

Now check the dot product at $t = \pi/2$:

$$\mathbf{v}\left(\frac{\pi}{2}\right) \cdot \mathbf{a}\left(\frac{\pi}{2}\right) = (-10\,\hat{\jmath}) \cdot (20\,\hat{\imath}) = 0.$$

So they are perpendicular at that instant.
In fact, for *all* t,

$$\begin{aligned}\mathbf{v}(t)\cdot\mathbf{a}(t) &= (-10\sin(2t))(-20\cos(2t)) + (10\cos(2t))(-20\sin(2t)) \\ &= 200\sin(2t)\cos(2t) - 200\sin(2t)\cos(2t) \\ &= 0.\end{aligned}$$

Thus $\mathbf{v}(t) \perp \mathbf{a}(t)$ for all time. This is characteristic of uniform circular motion: acceleration is purely centripetal (radially inward), velocity is purely tangential.

2. **Two particles in the plane.**

The position vectors are

$$\mathbf{r}_1(t) = (1+2t)\,\hat{\imath} + (3-t)\,\hat{\jmath}, \qquad \mathbf{r}_2(t) = (4-t)\,\hat{\imath} + (1+3t)\,\hat{\jmath}.$$

(a) *Velocities and accelerations.*
Differentiate each with respect to t.
For particle 1:

$$\mathbf{v}_1(t) = \frac{\mathrm{d}\mathbf{r}_1}{\mathrm{d}t} = 2\,\hat{\imath} + (-1)\,\hat{\jmath} = 2\,\hat{\imath} - \hat{\jmath},$$

$$\mathbf{a}_1(t) = \frac{\mathrm{d}\mathbf{v}_1}{\mathrm{d}t} = \mathbf{0}.$$

So particle 1 moves at constant velocity, no acceleration.
For particle 2:

$$\mathbf{v}_2(t) = \frac{\mathrm{d}\mathbf{r}_2}{\mathrm{d}t} = (-1)\,\hat{\imath} + 3\,\hat{\jmath} = -\hat{\imath} + 3\,\hat{\jmath},$$

$$\mathbf{a}_2(t) = \frac{\mathrm{d}\mathbf{v}_2}{\mathrm{d}t} = \mathbf{0}.$$

So particle 2 also moves at constant velocity, no acceleration.

(b) *Time of closest approach and minimum distance.*
The displacement from particle 2 to particle 1 is

$$\Delta\mathbf{r}(t) = \mathbf{r}_1(t) - \mathbf{r}_2(t).$$

Compute components:

$$\begin{aligned}\Delta x(t) &= (1+2t) - (4-t) = 1 + 2t - 4 + t = 3t - 3, \\ \Delta y(t) &= (3-t) - (1+3t) = 3 - t - 1 - 3t = 2 - 4t.\end{aligned}$$

Thus

$$\Delta\mathbf{r}(t) = (3t - 3)\,\hat{\imath} + (2 - 4t)\,\hat{\jmath}.$$

The square of the distance between them is

$$D^2(t) = \|\Delta\mathbf{r}(t)\|^2 = (3t - 3)^2 + (2 - 4t)^2.$$

Expand:

$$(3t - 3)^2 = 9t^2 - 18t + 9,$$
$$(2 - 4t)^2 = 16t^2 - 16t + 4.$$

So

$$D^2(t) = (9t^2 - 18t + 9) + (16t^2 - 16t + 4) = 25t^2 - 34t + 13.$$

To minimize the distance, we minimize $D^2(t)$ (same t since square root is monotonic). Take derivative:

$$\frac{\mathrm{d}D^2}{\mathrm{d}t} = 50t - 34.$$

Set to zero:

$$50t - 34 = 0 \quad \Rightarrow \quad t = \frac{34}{50} = \frac{17}{25} = 0.68 \text{ s}.$$

Now evaluate the minimum distance. First compute D^2 at $t = 17/25$:

$$D^2\left(\frac{17}{25}\right) = 25\left(\frac{17}{25}\right)^2 - 34\left(\frac{17}{25}\right) + 13.$$

Compute term by term:

$$\left(\frac{17}{25}\right)^2 = \frac{289}{625}, \quad 25 \cdot \frac{289}{625} = \frac{7225}{625} = 11.56.$$

Next:

$$34 \cdot \frac{17}{25} = \frac{578}{25} = 23.12.$$

So

$$D^2\left(\frac{17}{25}\right) = 11.56 - 23.12 + 13 = 1.44.$$

Therefore

$$D_{\min} = \sqrt{1.44} = 1.2 \text{ m}.$$

Summary: the two particles are closest at

$$t = \frac{17}{25} \text{ s} = 0.68 \text{ s}, \qquad D_{\min} = 1.2 \text{ m}.$$

Interpretation: since both have constant (but different) velocities, their relative motion is uniform, so the closest approach happens when the line between them is perpendicular to the relative velocity.